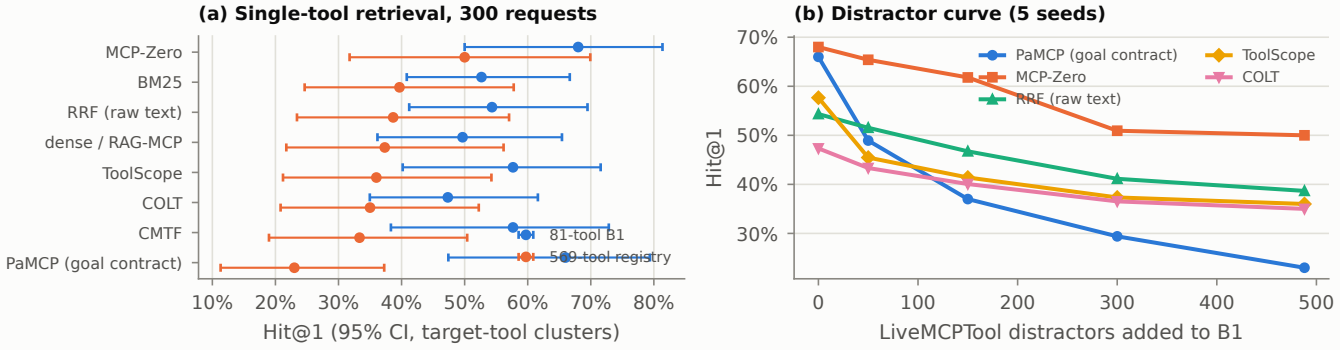
(a) Single-tool retrieval, 300 requests
MCP-Zero
BM25
RRF (raw text)
dense / RAG-MCP
ToolScope
COLT
CMTF
PaMCP (goal contract)
81-tool B1
509-tool registry
10%
20%
30%
40%
50%
60%
70%
80%
Hit@1 (95% CI, target-tool clusters)
(b) Distractor curve (5 seeds)
PaMCP (goal contract)
MCP-Zero
RRF (raw text)
ToolScope
COLT
Hit@1
70%
60%
50%
40%
30%
0
100
200
300
400
500
LiveMCPTool distractors added to B1